

IMDb MOVIE DATA ANALYSIS

Minor Project – Data Science

Python-Based Exploratory Data Analysis of IMDb Movie Data

Prepared using: Python, Pandas, NumPy, Matplotlib, Seaborn and Google Colab

Project Report

1. Introduction

The IMDb Movie Data Analysis project explores a real-world movie dataset using Python-based data analysis techniques. The dataset contains information about movie titles, genres, directors, actors, release years, runtime, IMDb ratings, votes, revenue and Metascore. The project applies data cleaning, exploratory analysis, filtering, grouping, sorting and visualization to discover useful patterns in the data.

The work began with the provided IMDb analysis notebook and dataset. The existing analysis was understood and executed, and additional analyses were added to extend the project with new insights about movie popularity, revenue, directors, actors and rating-related patterns.

2. Objectives

- To understand and explore the IMDb movie dataset.
- To identify and handle missing values in the dataset.
- To check the dataset for duplicate records.
- To analyze movie ratings, votes, revenue, runtime, genres, directors and actors.
- To identify highly rated, highly voted and high-revenue movies.
- To perform additional analysis beyond the original notebook.
- To visualize important results using charts and plots.
- To derive meaningful findings from the movie data.

3. Dataset Description

The dataset used in the notebook contains 1,000 movie records and 12 attributes. The original columns are Rank, Title, Genre, Description, Director, Actors, Year, Runtime (Minutes), Rating, Votes, Revenue (Millions) and Metascore.

Attribute	Description
Rank	Ranking of the movie
Title	Movie title
Genre	Genre or genre combination
Description	Short description of the movie
Director	Director of the movie
Actors	Main actors listed for the movie
Year	Release year
Runtime (Minutes)	Movie duration
Rating	IMDb user rating
Votes	Number of IMDb votes
Revenue (Millions)	Revenue in millions
Metascore	Critics' score

4. Technologies Used

Technology	Purpose
Python	Primary programming language for analysis
Pandas	Data loading, cleaning, filtering, grouping and sorting
NumPy	Numerical/data processing support
Matplotlib	Data visualization and charts
Seaborn	Statistical visualization
Google Colab	Notebook environment for executing Python

5. Methodology

1. **Data Loading** – The CSV dataset was loaded into a Pandas DataFrame in Google Colab.
2. **Data Exploration** – The first and last records, shape, data types and descriptive statistics were inspected.
3. **Missing-Value Analysis** – Missing values were identified using `isnull().sum()` and visualized with a heatmap.
4. **Data Cleaning** – Rows containing missing values were removed using `dropna()`, leaving 838 complete records.
5. **Duplicate Check** – The `duplicated()` method was used and no duplicate rows were found.
6. **Exploratory Analysis** – Filtering, grouping, sorting and aggregation were used to analyze movie characteristics.
7. **Additional Analysis** – New questions were added for votes, directors, genre ratings, revenue, recent highly-rated movies, actors and rating/revenue correlation.
8. **Visualization** – Bar charts, count plots, scatter plots and heatmaps were used to communicate findings.

6. Data Preprocessing Results

The original dataset contained 1,000 rows and 12 columns. The information check showed that Revenue (Millions) had 872 non-null values and Metascore had 936 non-null values. Therefore, 128 Revenue values and 64 Metascore values were missing.

Measure	Result
Original records	1,000
Original columns	12
Missing Revenue values	128
Missing Metascore values	64
Records after dropna()	838
Duplicate records	None found
Average rating after cleaning	6.8143
Average runtime after cleaning	114.64 minutes
Average revenue after cleaning	84.56 million
Average Metascore after cleaning	59.58

After cleaning, all remaining records had values for the analyzed columns. The cleaned dataset was then used for the subsequent analysis.

7. Original Analysis

The provided notebook includes a sequence of exploratory analyses. These include checking long movies, visualizing votes and revenue by year, calculating average rating by director, finding longest movies, counting movies per year, identifying the highest-revenue movie, listing the highest-rated movies, finding the highest-revenue movies, calculating year-wise average ratings, examining rating versus revenue, classifying movies by rating, and counting Action movies.

Important results from the original analysis

- Three movies have runtime of at least 180 minutes: The Wolf of Wall Street, The Hateful Eight and La vie d'Adèle.
- Christopher Nolan has the highest average director rating in the notebook output at 8.68.
- The Dark Knight is the highest-rated movie in the cleaned dataset with a rating of 9.0.
- Star Wars: Episode VII - The Force Awakens has the highest recorded revenue at 936.63 million.
- The cleaned dataset contains 277 Action movies.
- The highest year-wise average rating shown is 7.1439 for 2006.

Top 10 Highest-Rated Movies

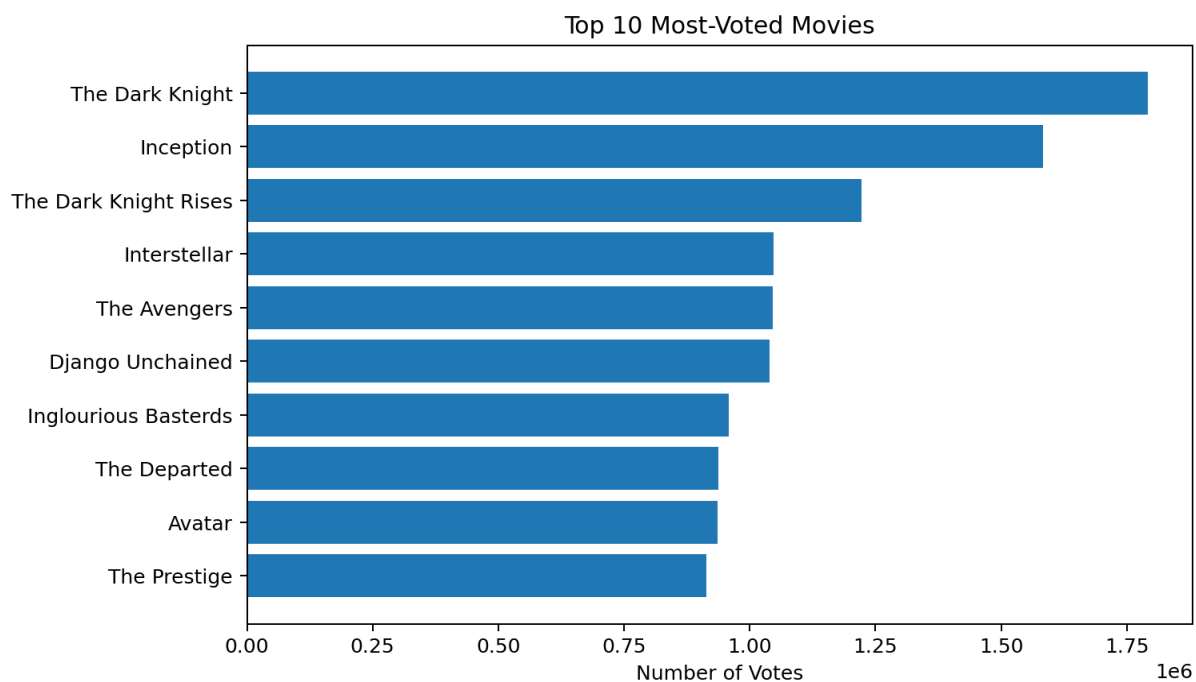
Movie	Rating	Director
The Dark Knight	9.0	Christopher Nolan
Inception	8.8	Christopher Nolan
Interstellar	8.6	Christopher Nolan
Kimi no na wa	8.6	Makoto Shinkai
The Intouchables	8.6	Olivier Nakache
The Prestige	8.5	Christopher Nolan
The Departed	8.5	Martin Scorsese
The Dark Knight Rises	8.5	Christopher Nolan
Whiplash	8.5	Damien Chazelle
The Lives of Others	8.5	Florian Henckel von Donnersmarck

8. Additional Analysis – Individual Contribution

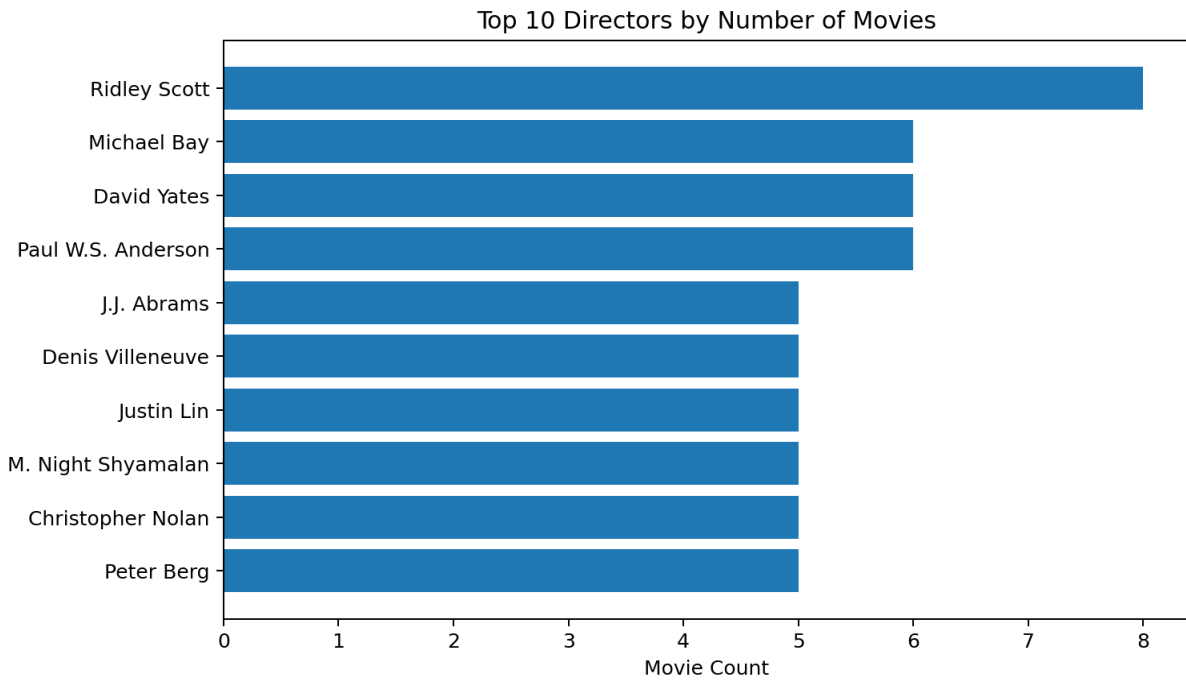
Additional analyses were added to extend the original notebook. The new work uses Pandas operations such as `nlargest()`, `value_counts()`, `groupby()`, filtering with multiple conditions, string splitting/exploding, and correlation analysis.

8.1 Top 10 Movies by Number of Votes

Movie	Votes	Rating
The Dark Knight	1,791,916	9.0
Inception	1,583,625	8.8
The Dark Knight Rises	1,222,645	8.5
Interstellar	1,047,747	8.6
The Avengers	1,045,588	8.1
Django Unchained	1,039,115	8.4
Inglourious Basterds	959,065	8.3
The Departed	937,414	8.5
Avatar	935,408	7.8
The Prestige	913,152	8.5



8.2 Top 10 Directors by Number of Movies

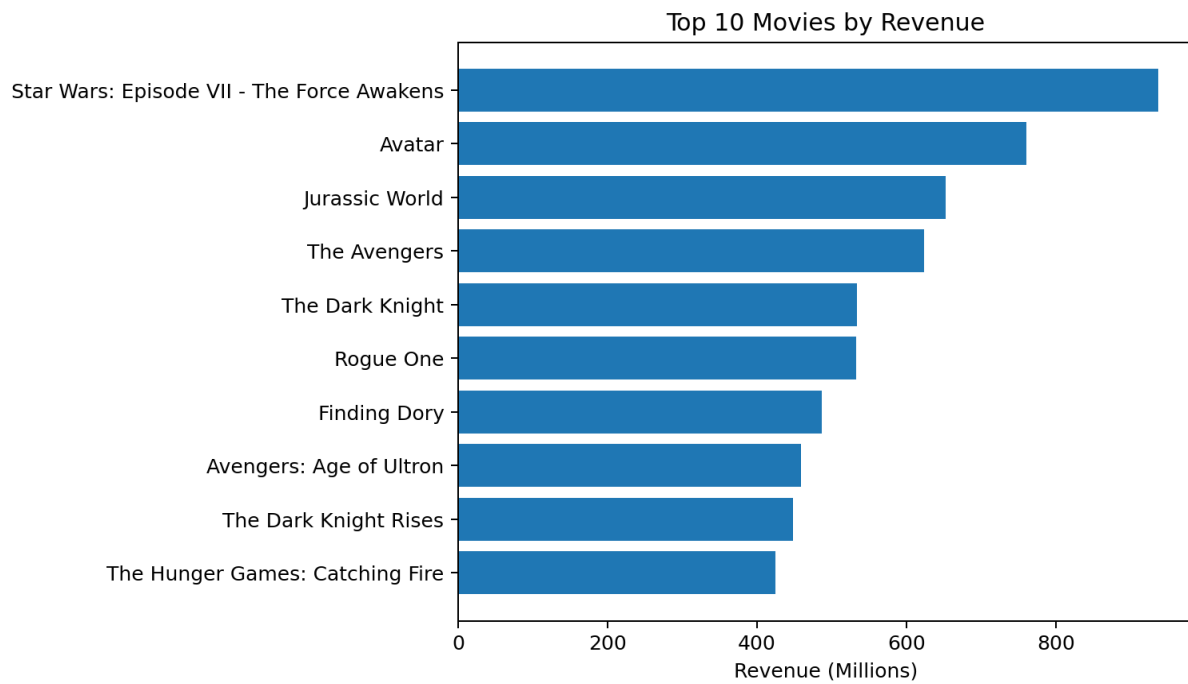


Ridley Scott appears most frequently among the directors in the cleaned dataset, with 8 movies. Michael Bay, David Yates and Paul W.S. Anderson each appear with 6 movies.

8.3 Average IMDb Rating by Genre Combination

The analysis grouped the dataset by the exact text stored in the Genre column. The highest average rating shown is 8.6 for Animation,Drama,Fantasy. Because the source Genre field contains combined genre strings, this analysis represents genre combinations rather than individual genres.

8.4 Top 10 Movies by Revenue



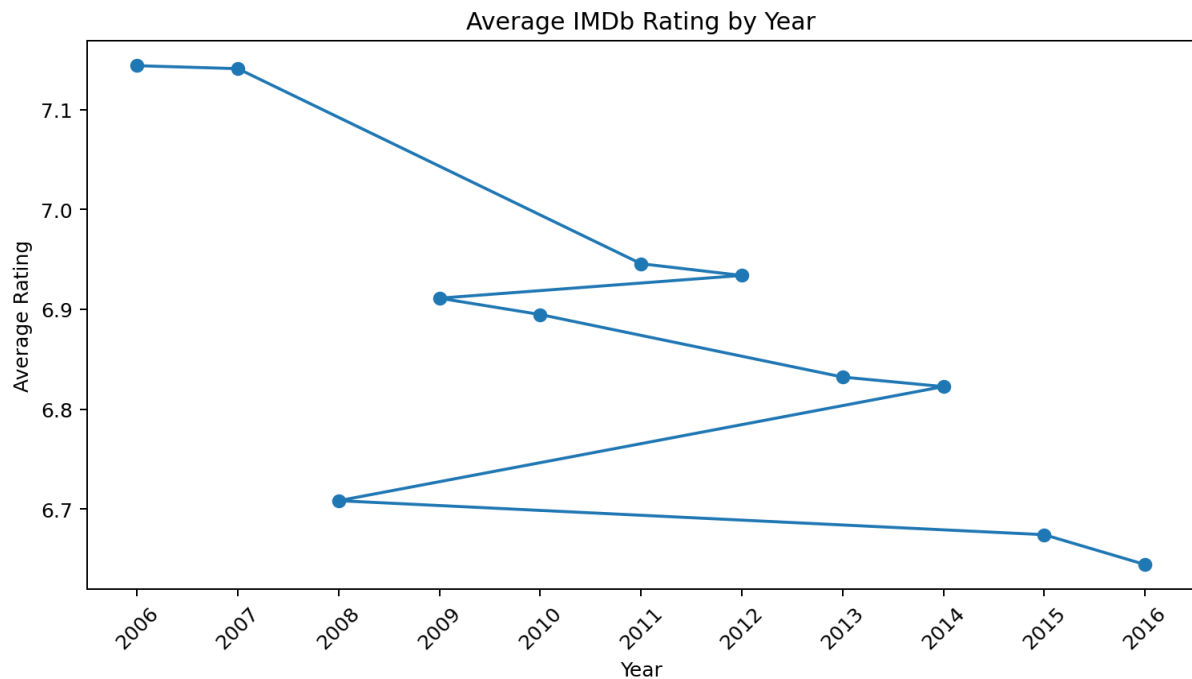
Star Wars: Episode VII - The Force Awakens leads the cleaned dataset with revenue of 936.63 million, followed by Avatar at 760.51 million and Jurassic World at 652.18 million.

8.5 Movies Released After 2010 with Rating Above 8

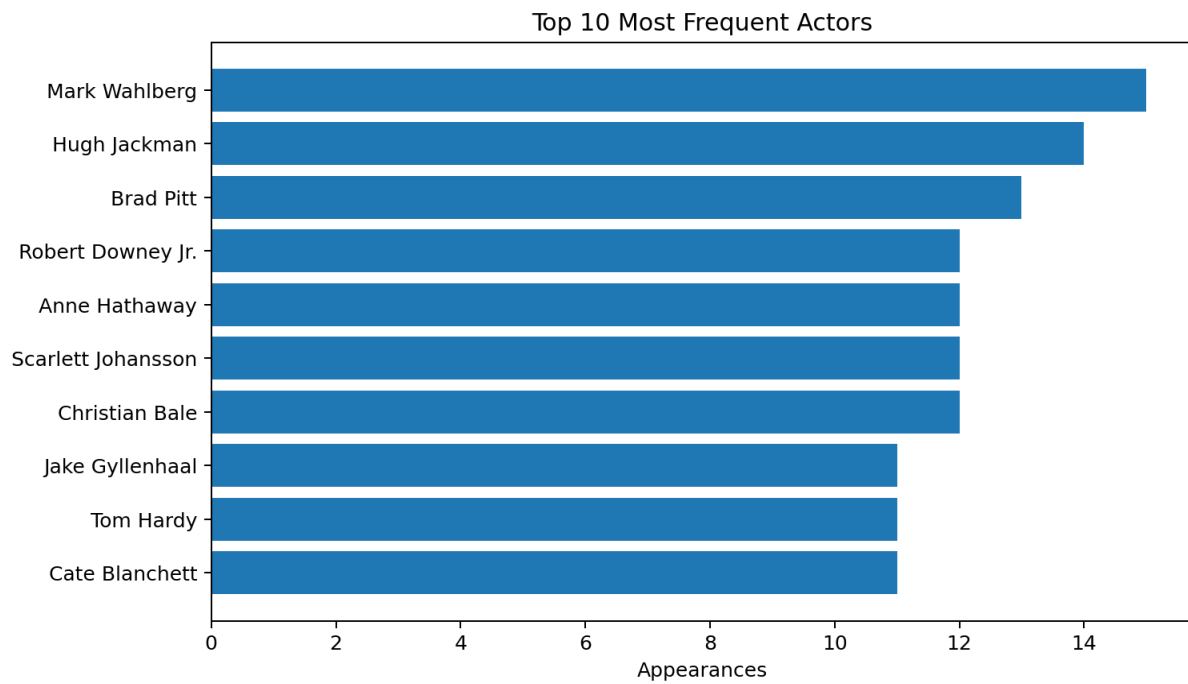
The filtering analysis identified 32 movies satisfying both conditions in the notebook output. The highest-rated entries in this filtered group are Interstellar, Kimi no na wa and The Intouchables, each with a rating of 8.6.

8.6 Relationship Between IMDb Rating and Revenue

The Pearson correlation calculated in the notebook is **0.217**. This indicates a weak positive linear relationship between rating and revenue in this cleaned dataset. The scatter plot provides a visual representation of the same relationship. Correlation should not be interpreted as proof that rating causes revenue.



8.7 Top 10 Most Frequent Actors



Mark Wahlberg appears most frequently in the Actors field, with 15 appearances. Hugh Jackman appears 14 times and Brad Pitt 13 times. The actor analysis was performed by splitting the comma-separated Actors field and counting the resulting names.

9. Key Findings

- The dataset started with 1,000 movie records and 12 attributes.
- Missing values were present in Revenue (128) and Metascore (64).
- After removing rows with missing values, 838 records remained.
- No duplicate records were found.
- The average IMDb rating of the cleaned dataset is approximately 6.81.
- The Dark Knight has the highest rating in the analyzed dataset at 9.0.
- Star Wars: Episode VII - The Force Awakens has the highest recorded revenue at 936.63 million.
- The Dark Knight has the highest number of votes at 1,791,916.
- Ridley Scott has the largest number of director entries in the additional director-frequency analysis.
- Mark Wahlberg is the most frequent actor in the additional actor-frequency analysis.
- The rating/revenue correlation is 0.217, indicating a weak positive linear relationship.

10. Conclusion

The IMDb Movie Data Analysis project demonstrates a complete exploratory data-analysis workflow. The dataset was loaded, inspected, cleaned and analyzed using Python libraries. The original notebook provided the foundation, while the additional analyses extended the project to include movie popularity, director frequency, genre-rating combinations, revenue, recent highly-rated movies, actor frequency and rating/revenue correlation.

The project shows how structured movie data can be transformed into useful information through filtering, sorting, grouping, aggregation and visualization. The findings also demonstrate that different movie attributes provide different perspectives on popularity and performance.

11. Future Scope

- Build an interactive dashboard using Tableau or Power BI.
- Perform deeper statistical analysis of rating, revenue and votes.
- Analyze individual genres by splitting combined genre values.
- Study trends across directors, actors and production periods.
- Develop predictive models for rating or revenue as a future extension.

12. References

- IMDb Movie dataset and the provided project notebook repository.
- Python documentation and standard Pandas, NumPy, Matplotlib and Seaborn usage.